Machine learning significantly improves neoantigen-HLA predictions utilizing > 26,000 data points from the PACTImmune™ Database

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Introduction

PACT Pharma has developed a state-of-the-art approach to validate predicted neoepitopes (neoEs) and their cognate T cell receptors (neoTCRs) by capturing neoepitope-specific T cells from peripheral blood. This neoTCR discovery and validation process is being applied in clinical trial (NCT03970382) evaluating personalized neoTCR-T cell therapy to treat patients across eight solid tumor types. Extensive pre-, on- and post-treatment data related to this trial has been accumulated in the PACTImmune™ Database (**PIDB**) which represents a growing data asset for patient-specific tumor immunogenicity in solid tumors. Here we present a specific use case of applying machine learning (**ML**) to significantly improve neoE-HLA predictions and further model anticipated improvements of TCR capture as a direct consequence.

Methods

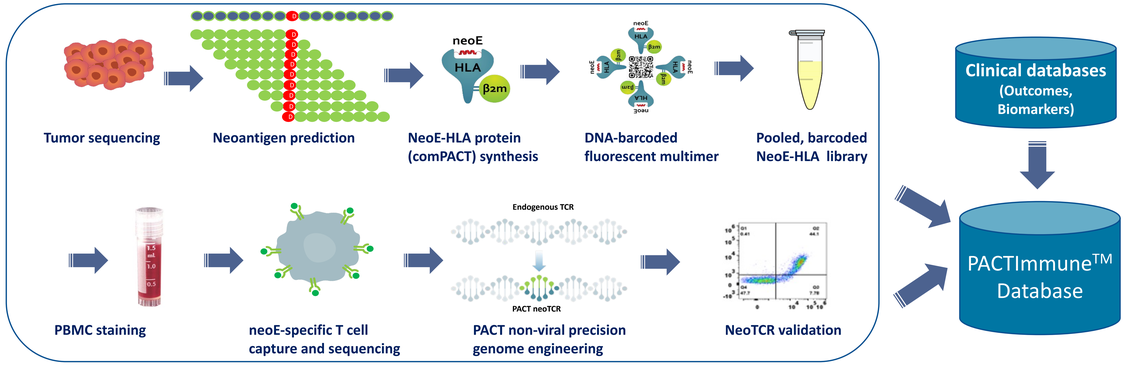


Figure 1. Overview of the PACT process for isolating and validating TCRs from patient samples and the types of data captured in the PIDB.

PACT has developed capabilities for high-throughput manufacturing of single polypeptide (comPACT protein) which consists of the predicted neoE peptide together with beta-2-microglobulin and the HLA heavy chain. comPACT molecules are considered successfully produced when protein yields reach concentrations >1uM. We applied ML to learn patterns that are predictive of neoE-HLAs that can be successfully produced as comPACTs. Data used for this study consisted of >26,000 neoE-HLA predictions for 62 different HLA alleles (see Model Training section for details). We additionally performed experimental prospective validation of the models. To do this, 603 neoE-HLAs (from 7 previously unseen cancer samples) were selected using netMHCpan4.1 and the newly trained models and comPACT success rate was compared.

PIDB neoE-HLA data

PACTImmune™ Database	
# Samples evaluated	113
# neoE-HLA predictions	26002
# HLA alleles	62

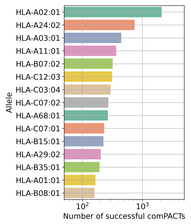


Figure 2. **(left)** PACTImmune Database summary **(right)** Top 15 HLA alleles with the highest number of successful comPACTs.

Amino acid descriptors

no.	descriptor variables
1	molecular weight (g/mol)
2	TLC % migration on silica gel, ethanol/water (70/30)[a]
3	TLC, silica gel, 1-butanol/acetic acid/water (40/10/10)
4	TLC, silica gel, phenol/water (75/25)
5	TLC, silica gel, butanone/pyridine/acetic acid/water (70/15/2/15)
6	TLC, cellulose, ethanol/water (70/30)
7	TLC, cellulose, pyridine/isoamyl alcohol/water (35/30/30)
8	TLC, kieselguhr, butanone/water/phenol/acetone/ethanol (1/1)
9	side chain van der Waals volume (cm^3/mol)
10	NMR α-proton shift at pD = 2 (ppm)
11	NMR α-proton shift at pD = 7 (ppm)
12	NMR α-proton shift at pD = 12.5 (ppm)
13	^{18}log(octanol/water) partition coefficient
14	energy of highest occupied molecular orbital (eV)
15	energy of lowest unoccupied molecular orbital (eV)
16	heat of formation (kcal)
17	α-polarizability (Å^3)
18	absolute electronegativity (eV)
19	absolute hardness (eV)
20	total accessible molecular surface area (log Å^2)
21	polar accessible molecular surface area (log Å^2)
22	nonpolar accessible molecular surface area (log Å^2)
23	number of hydrogen bond donors
24	number of hydrogen bond acceptors
25	indicator of positive charge in side chain
26	indicator of negative charge in side chain

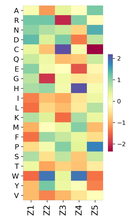


Figure 3. **(left)** The 26 descriptors based on which the Z-Scales were derived.[1] **(right)** Z-scales values for the 20 amino acids.

Peptides of multiple length can use padding to simplify the number of models

To incorporate peptides of multiple lengths in the same ML model, a pseudo-amino acid (X) was added at non-anchor residue positions using a similar approach that is used in the literature[2]. All peptide of length 8-10 were padded such that the resulting padded peptides were of length 11.

Peptide Length	Before padding	After padding
8-mer	AAAAAAAA	AAAAXXXAAAA
9-mer	LLLLLLLLL	LLLLXLXLLLL
10-mer	YYYYYYYYYY	YYYYXYYYYYY
11-mer	IIIIIIIIIII	IIIIIIIIIII

Table 1. Addition of pseudo amino acid X at specific positions to make all peptides of length 11.

Model training

- For each allele, data was first split into **training** and **testing** data using a 70/30 stratified split.
- XGBoost and Random Forest models were trained on training data and model hyperparameters were tuned using 5-fold cross validation (5xCV).
- Performance measured using the area under the receiver operating characteristic curve (AUROC).
- For each HLA allele, the best performing model in the 5xCV was selected and its performance was determined using the held-out test data.

Prospective validation

Allele	Sample ID	# Expressed mutations	Attempts
HLA-A03:01	1	243	55
HLA-A02:01	2	15	72
	3	24	70
HLA-C07:02	1	243	59
	2	15	78
HLA-C07:01	4	78	71
HLA-C03:04	5	120	71
HLA-B15:01	6	328	69
HLA-B35:01	7	44	58

Table 2. Additional samples used for prospective validation of the models. The column 'Attempts' shows the predetermined number of neoE-HLAs for comPACT production that were selected using netMHCpan4.1 and models based on PACTImmune Database. In total, 603 neoE-HLAs (from 7 previously unseen cancer samples) were selected using netMHCpan4.1 and the newly trained models and comPACT success rate was compared

Results

The mean AUC for the 5xCV of the selected models ranged from 0.75 to 0.86 depending upon the HLA allele (SD <0.05 for every model). The AUC on the test data ranged from 0.75 to 0.92 (median = 0.85). Prospective validation resulted on average in a 22% higher success rate (range 11%-39%) using the new models as compared to the netMHCpan4.1 predictions. This is expected to result in increased capture of neoepitope-specific CD8+ T cells as the PIDB indicates that 3.2% of the successful comPACTs result in validated neoTCRs.

Peptide sequences relate to comPACT success in an HLA allele- and an amino acid position-specific manner

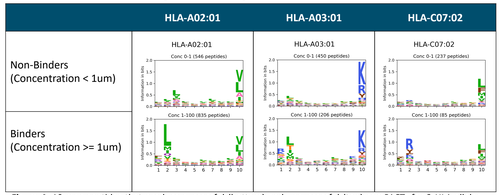


Figure 4. 10-mer peptides that produce successful (bottom) and unsuccessful (top) comPACTs for 3 HLA alleles show amino acid differences at key positions.

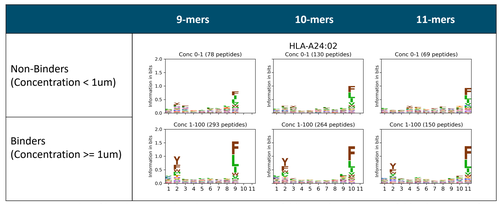


Figure 5. Peptides of length 9, 10, and 11 that produce successful (bottom) and unsuccessful (top) comPACTs for HLA-A24:02 show amino acid differences at key positions.

Combining data from different peptide lengths improves model performance

Peptide length	9-mer		10-mer		11-mer	
Case	A	B	A	B	A	B
Precision	0.82	0.81	0.81	0.84	0.77	0.80
Recall	0.89	0.91	0.81	0.79	0.81	0.83
AUROC	0.75	0.77	0.84	0.86	0.83	0.85

Table 3. Performance comparison for one HLA allele when the models were trained using data from same lengths (Case A) and data from all lengths (Case B). Results shown here are the mean metrics for 5 splits of data for each peptide length. For one split, for Case A, the model was trained using the remaining 4 splits of the same peptide length, whereas for Case B, the model was trained using 4 splits from the same peptide length and all data from other peptide lengths.

PACT ML Outperforms netMHCpan4.1 Elution Rank & Binding Affinity Predictions

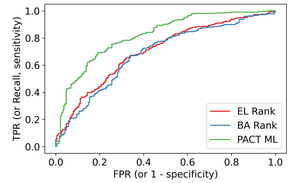


Figure 6. Comparison of ROC curves for 3 neoE-HLA ranking strategies, i.e., elution rank (netMHCpan4.1), binding affinity rank (netMHCpan4.1) and PACT ML scores, for HLA-A02:01.

comPACT success rates were on average 22% higher during prospective validation

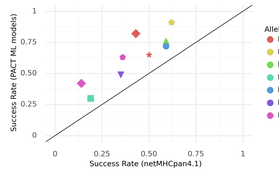


Figure 8. Comparison of the comPACT production success rates when the neoE-HLAs were selected using netMHCpan4.1 and the ML models based on PACTImmune Database. Each dot in the plot denotes a combination of a patient sample (shape) and an HLA allele (color).

AUROC on held out test data ranged between 0.75-0.92 for the selected 11 HLA alleles

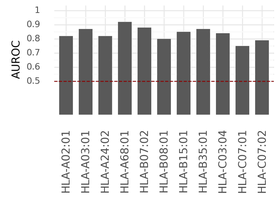


Figure 7. Performance of the selected models on test data (30% of data was previously unseen) in terms of area under the ROC curve (AUROC).

PACT ML is widely applicable to patients

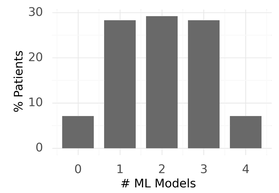


Figure 9. Percentage of patient samples versus the number of ML models applicable for the 11 selected HLAs. The x-axis shows the number of alleles for a sample that are in the group of 11 selected HLA alleles with ML models.

Next Steps – Preliminary Results

Rationally grouping alleles helps rescue underrepresented alleles

Allele	HLA-C07:02		HLA-C07:01		HLA-C06:02
# comPACTs	727		540		292
Case	A	B	A	B	Train on other 2
AUROC	0.76	0.75	0.79	0.78	0.80

Table 4. Performance comparison when the models were trained using data from same allele (Case A) and data from "similar" alleles (Case B). Results shown here are the mean metrics for 5 splits of data for each allele. For one split, for Case A, the model was trained using the remaining 4 splits of the same allele, whereas for Case B, the model was trained using 4 splits from the same allele and all data from other "similar" alleles.

Exploration of alternative models

	Random Forest	XGBoost
HLA-A02:01	0.82	0.84
HLA-B07:02	0.88	0.91
HLA-A03:01	0.87	0.87
HLA-C07:02	0.79	0.80
HLA-A24:02	0.82	0.82
HLA-C03:04	0.84	0.84

Table 5. Performance comparison (AUROC) of RF and XGBoost models on the test data.

Conclusions

- The PACTImmune Database of neoE predictions, neoTCR validations and clinical data represents a significant & growing dataset for patient-specific tumor immunogenicity in solid cancers.
- PACTImmune Database-based machine learning improved predictions of neoE-HLAs and translates to a significant increase in successful manufacturing of comPACT molecules.
- The increase in successfully manufactured comPACT molecules is expected to result in an increase in neoE-specific CD8+ T cell capture and actionable TCRs per patient.
- Future improvements in heuristics for epitope selection & prioritization are underway.

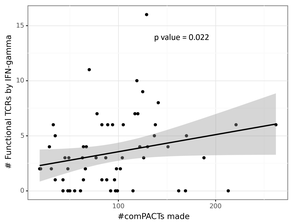


Figure 10. Improved comPACT production success rates from ML is predicted to lead to more TCR options per patient. Plot shows retrospective PIDB data of the number of functional TCRs determined by IFN-γ expression verses the number of successful comPACTs made for that patient. A significant positive correlation (P=0.022) is found between the two.

References

1 Sandberg, M., Eriksson, L., Jonsson, J., Sjöström, M., & Wold, S. (1998). New Chemical Descriptors Relevant for the Design of Biologically Active Peptides. A Multivariate Characterization of 87 Amino Acids. Journal of Medicinal Chemistry, 41(14). https://doi.org/10.1021/jm9700575

2 Zhao, T., Cheng, L., Zang, T., & Hu, Y. (2019). Peptide-Major Histocompatibility Complex Class I Binding Prediction Based on Deep Learning With Novel Feature. Frontiers in Genetics, 10. https://doi.org/10.3389/fgene.2019.01191